BLOCK 2

Module 5: AI in Project Optimisation, Innovation and Ethics

Unit 1: Time Series And Practical Applications

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- R&D Telecom Analytics Engineer [*AXON Networks, Madrid*]

Lecturer of:

- **M5 U1: Time Series And Practical Applications** *[AI in Project Optimisation, Innovation and Ethics]*

Class Learning Path: From Data to Prediction

- **Time Series Fundamentals:** Understanding what time series data is and why temporal order is essential.
- **Time Series Components:** Identifying trend, seasonality, and noise through visualization and decomposition.
- **Temporal Dependence:** Using autocorrelation analysis to understand how past values influence future behavior.
- **Model Evaluation:** Time-aware train-test splitting and the use of metrics to assess forecast quality.
- **Forecasting Models:** Applying classical (ARIMA) and modern (Prophet) methods for time series prediction. Conceptual understanding of LSTM (Optional).

- **Hands-On with Python:**
  - Manipulation and visualization of time series data
  - Training and evaluating forecasting models
  - Visual interpretation of forecasts and uncertainty

Learning Objectives And Practical Vision

From Time Series Data to Decisions

Forecasting is not only about predicting future values, but about understanding patterns to support real-world decisions.

Models as Decision-Support Tools

Forecasting models help guide decisions, they do not replace engineering judgment.

Learning by Doing Approach

Theory is introduced only when needed and immediately applied through hands-on Python examples

Group Activity: Time Series in Practice

Group Activity (FMP groups)
End-to-end time series analysis using Python on an electricity consumption dataset.

Core Exercises (5 × 2 points = 10 points)

1. Data Cleaning & Preprocessing
2. Multi-Scale Visualization
3. Seasonality Analysis
4. Statistical Analysis (ACF / PACF)
5. Forecasting with Prophet
6. *Bonus (Optional, +2 points):* Alternative forecasting with ARIMA or LSTM

Assessment Rules:

- *Scoring:* 2 points per exercise
- *Maximum grade:* 10 points
- *Bonus = wildcard:* You can skip or fail one exercise and still reach the maximum grade

Practical Notes

- *Dataset:* Electricity consumption time series. Very similar to the one used in class. Similar data type and structure, small differences only
- *In-class preparation:* We will solve the same type of exercises step by step during class
- *Deliverable:*
  Google Colab notebook with executable Python code + brief comments for each task

Class Calendar

- **23/02/2026: M5 U1 LS1:**

❑ From Chaos to Clarity: Time Series Visualization and Understanding

- **25/02/2026: M5 U1 LS2:**

❑ Predicting the Future: Time Series Forecasting

M5 U1 LS1

From Chaos to Clarity: Time Series Visualization and Understanding

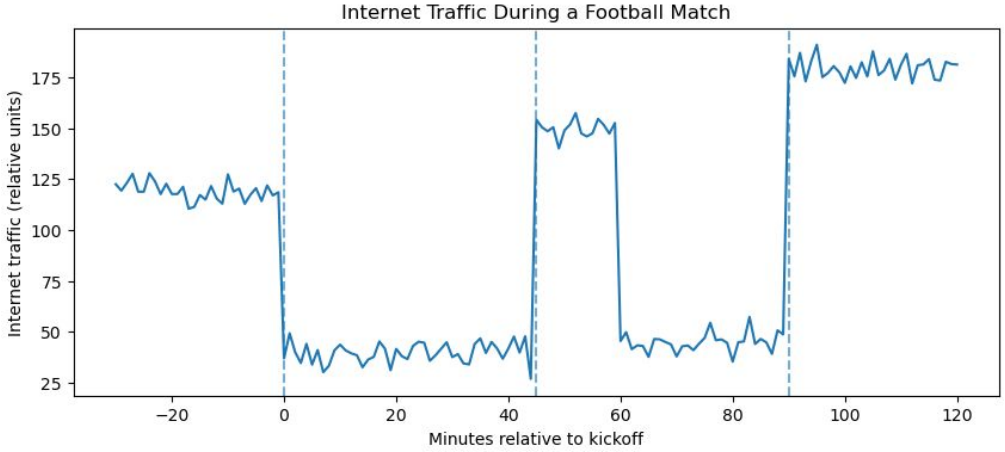
Time Series Fundamentals

What Is a Time Series?

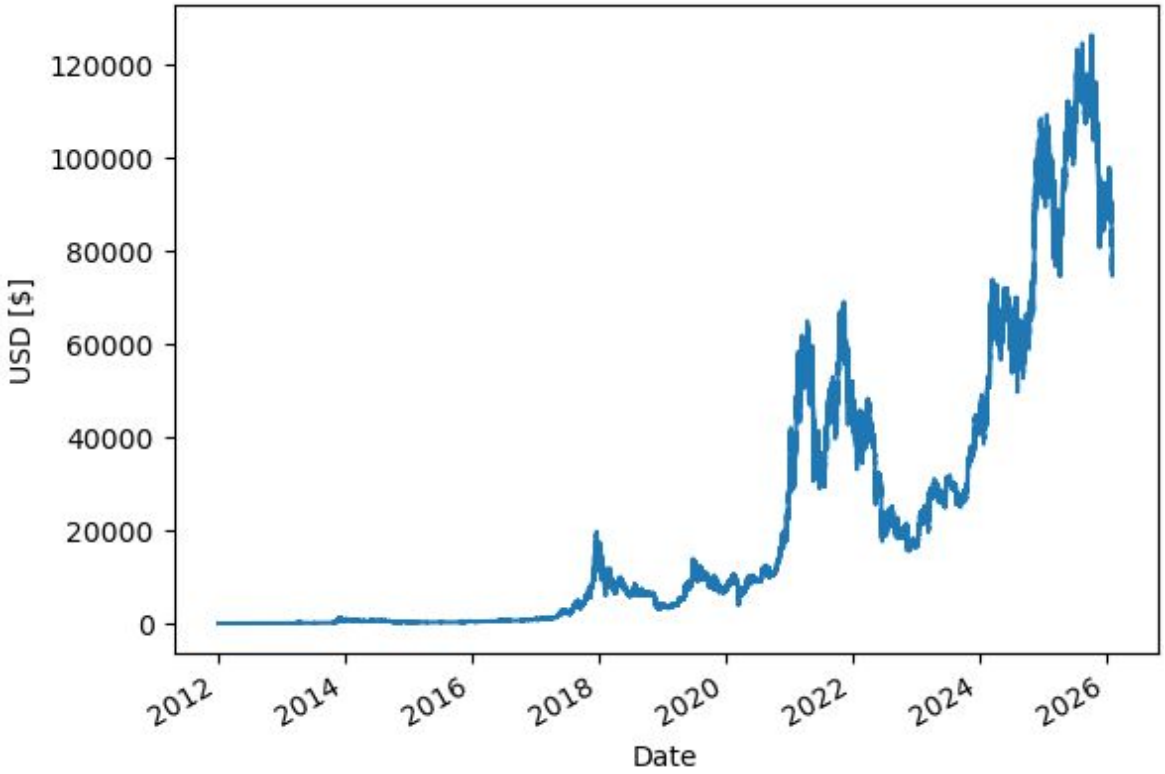
A time series...

- **Formal:** ... is a sequence of observations collected over time, where the order of the data matters.
- **Informal: ...** represents how a quantity evolves over time.

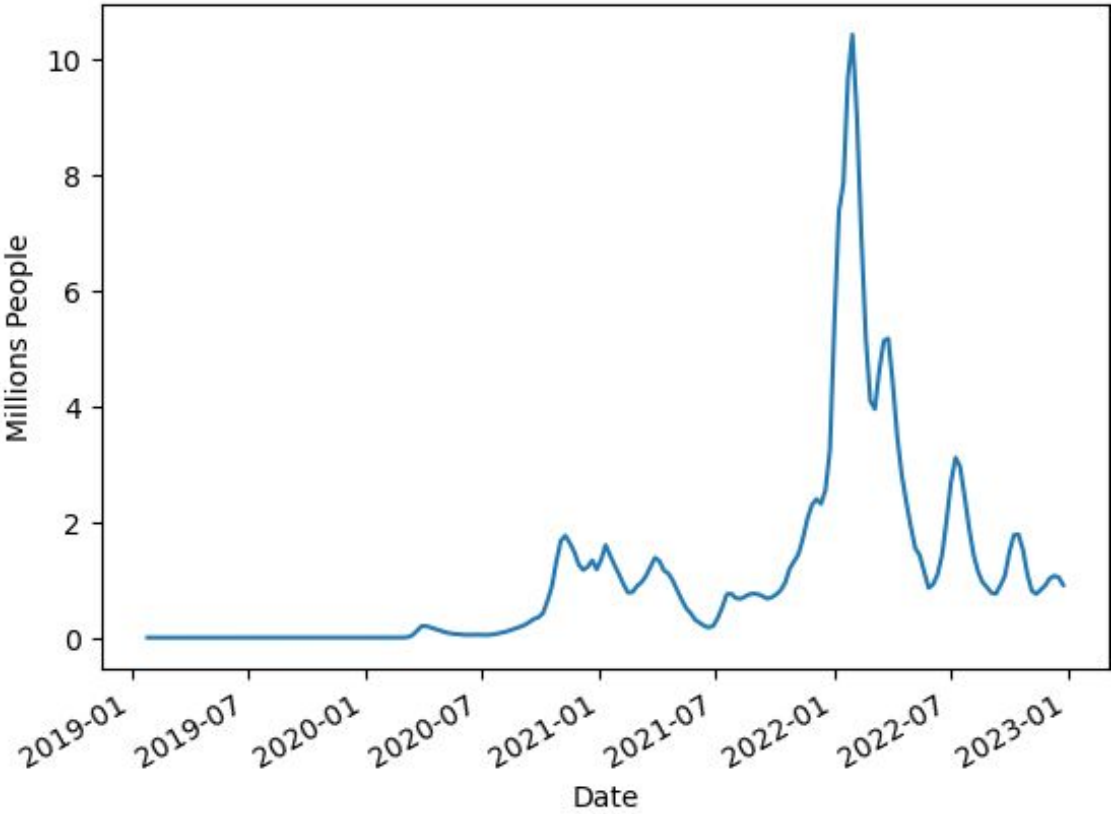
Key idea: Time gives structure to the data, past values influence the future. If we shuffle the data, it becomes meaningless.



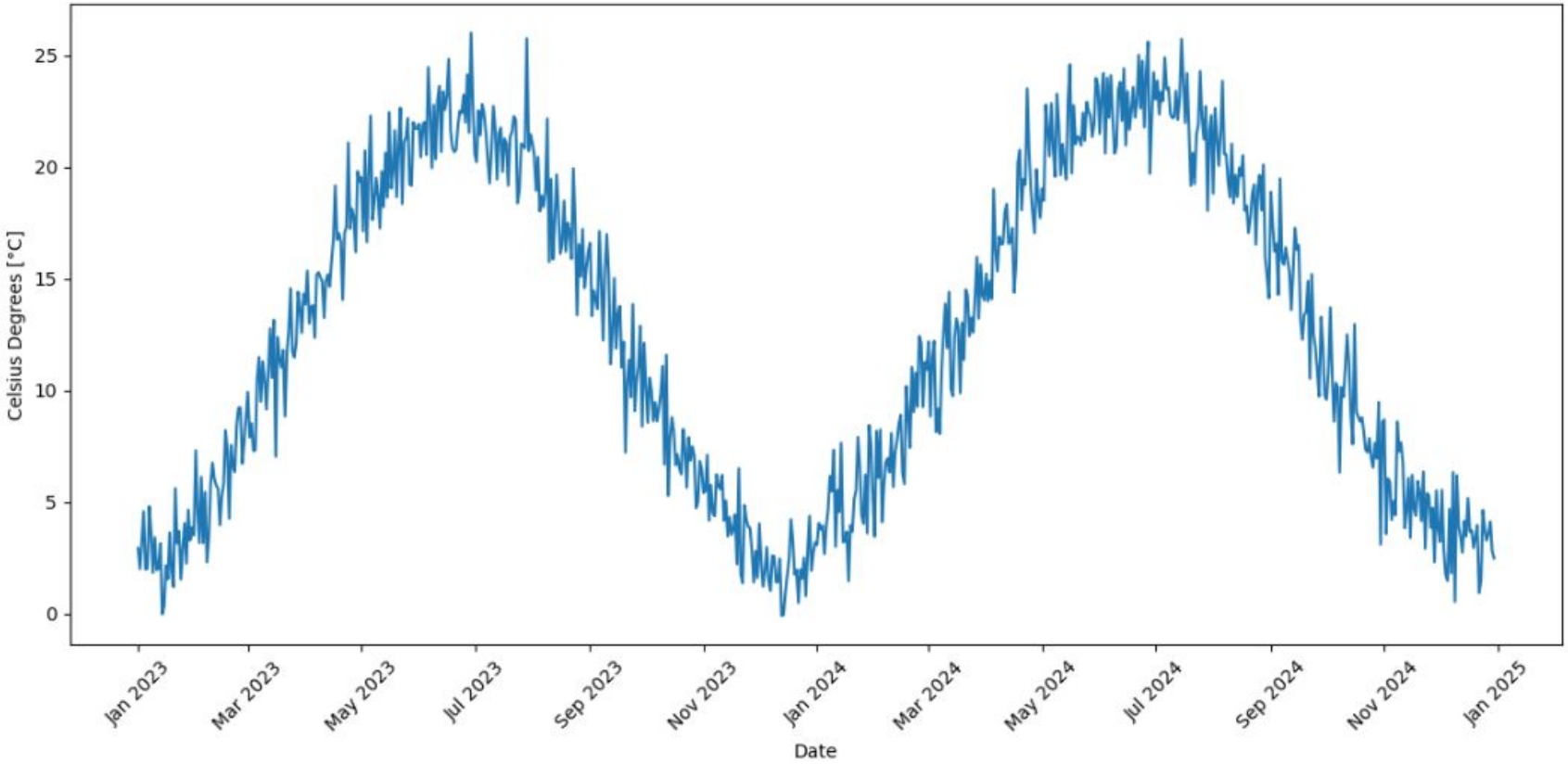
Famous Time Series - Guess the Story (I)



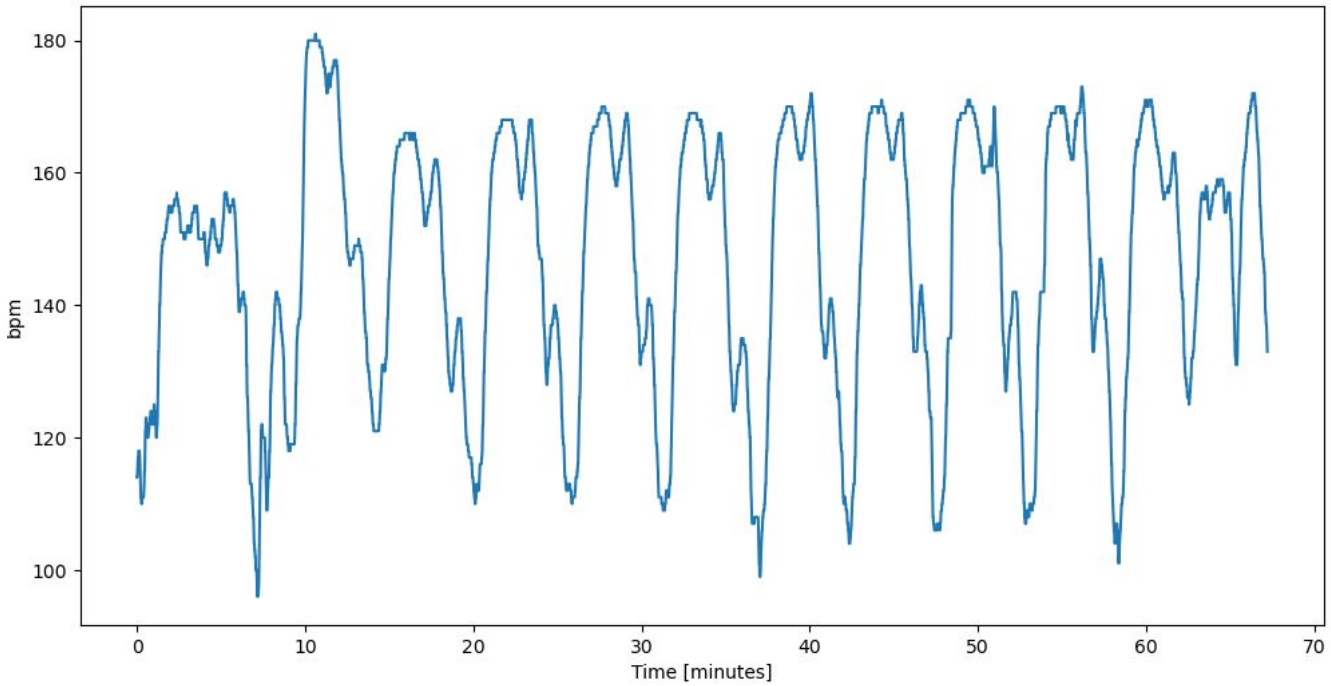
Famous Time Series - Guess the Story (II)



Famous Time Series - Guess the Story (III)

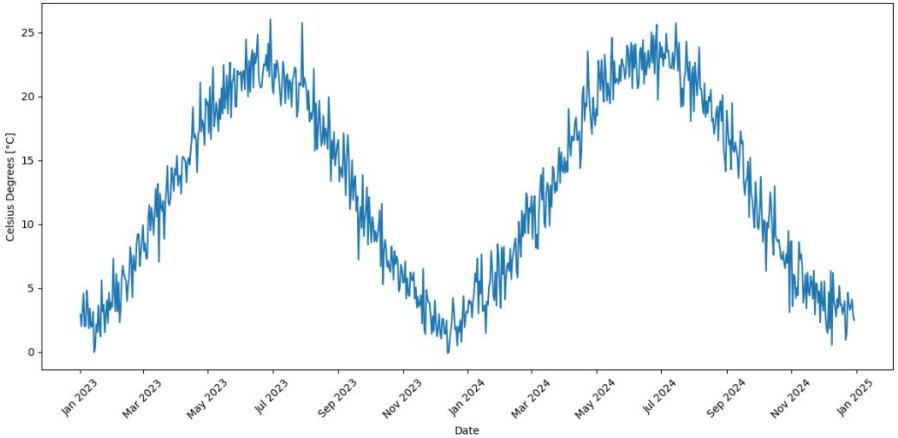


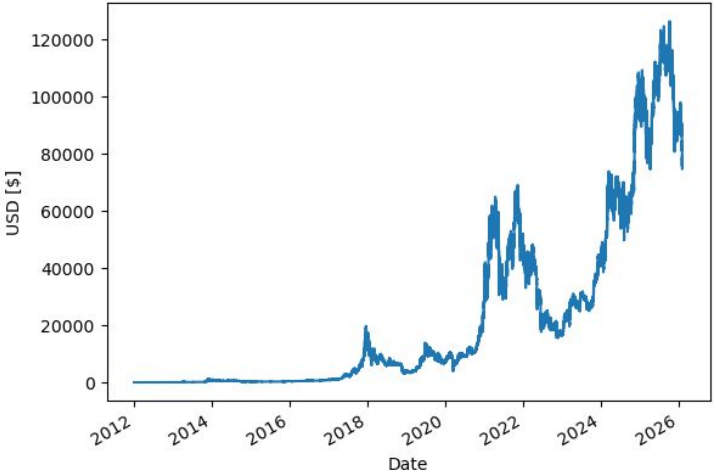
Famous Time Series - Guess the Story (IV)

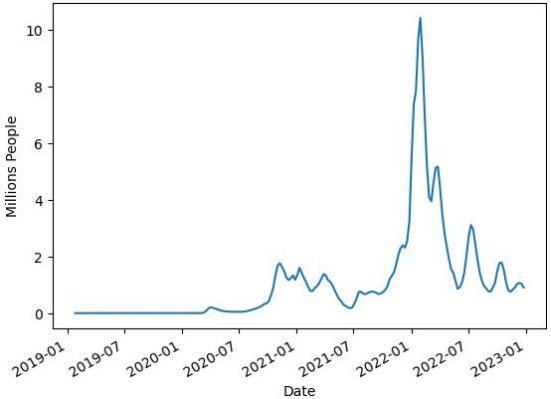


Famous Time Series: What We Just Saw

- Evolution over time
- Patterns and irregularities
- Clear seasonal patterns
- Increasing trends
- ...



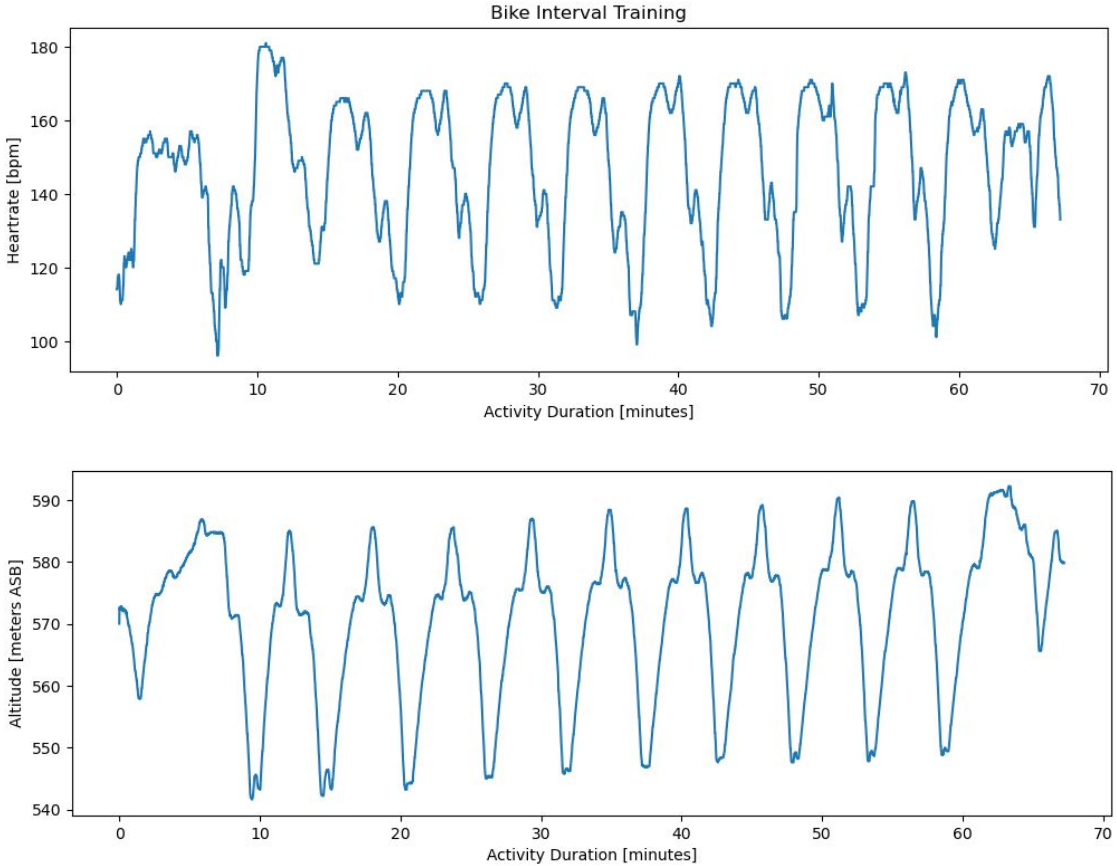




Correlated Time Series: A First Intuition

Correlation measures how two time series change in relation to each other.

- **Positive correlation**: When one goes up, the other tends to go up.
- **Negative correlation:** When one goes up, the other tends to go down.
- **No correlation:** Changes in one don't tell you anything about the other.



Heart rate rises as the altitude increases during interval training. This shows a positive correlation between heart rate and climbing intensity

\ What Is Forecasting? What Are We Really Doing? Why Does It Matter?

What is Forecasting?

- Use past data to estimate future values
- Assume patterns partially persist
- Produce predictions + uncertainty

What are we really doing?

- Learning patterns from historical data
- Quantifying uncertainty about the future
- Building models that approximate reality

Why Does It Matter?

- Decisions concern the future
- **Goal:** be less wrong than guessing

📊 **Data**
Raw measurements over time
↓
🔍 **Patterns**
Trend, seasonality, correlations
↓
🤖 **Model**
Mathematical approximation of reality
↓
🔮 **Forecast**
Future values + uncertainty
↓
✅ **Decisions**
Planning, design, operations

All models are wrong, but some are useful
[George Box]

Time Series in Civil Engineering & Architecture

🏗️ Project Cost Forecasting

- Predict final project cost from weekly/monthly spending
- Detect budget overruns early
- Support financial planning
- ...

🏢 Building Operations Forecasting

- Energy consumption forecasting for HVAC sizing
- Peak load prediction
- Future maintenance needs
- ...

⏱️ Schedule & Delay Forecasting

- Forecast project completion dates
- Predict delays based on historical progress curves
- Identify critical phases likely to slip
- ...

🌍 Infrastructure Demand Forecasting

- Traffic volume prediction
- Water and energy demand forecasting
- Usage evolution over time
- ...

Time Series Components

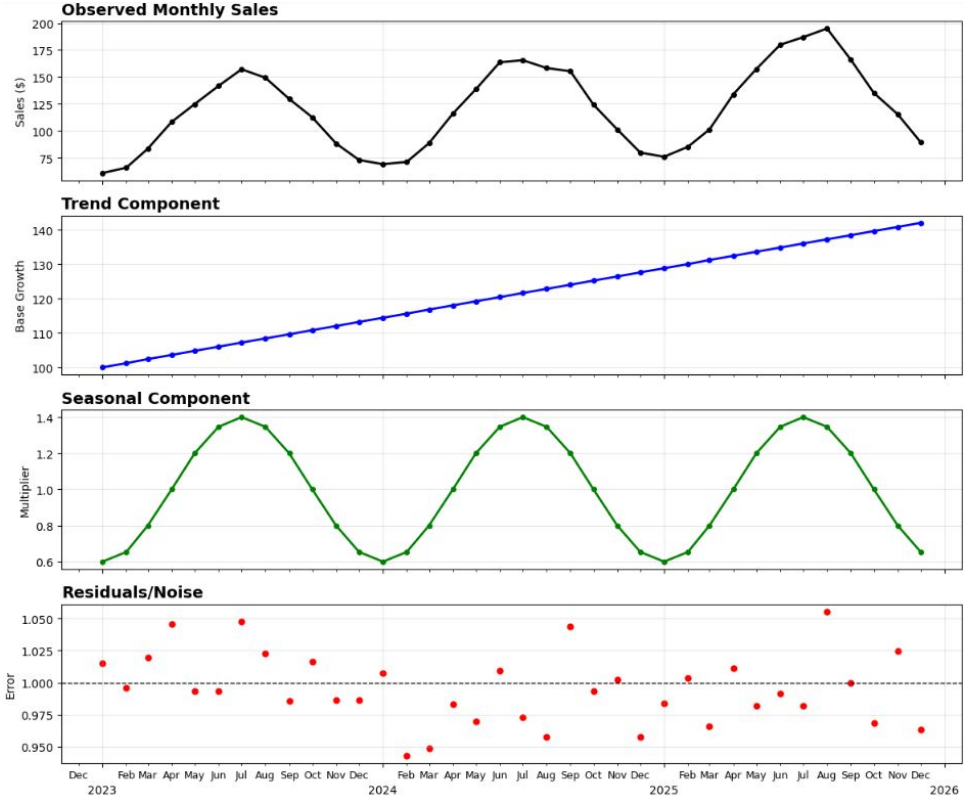
What Is Inside a Time Series? Seasonal Decomposition

A time series is usually a combination of:

- ***Trend:*** Long-term direction (increase, decrease, or stable)
- ***Seasonality:*** Regular repeating patterns (daily, weekly, yearly)
- ***Residual (Noise):*** Random fluctuations we cannot easily explain

Why Decompose?

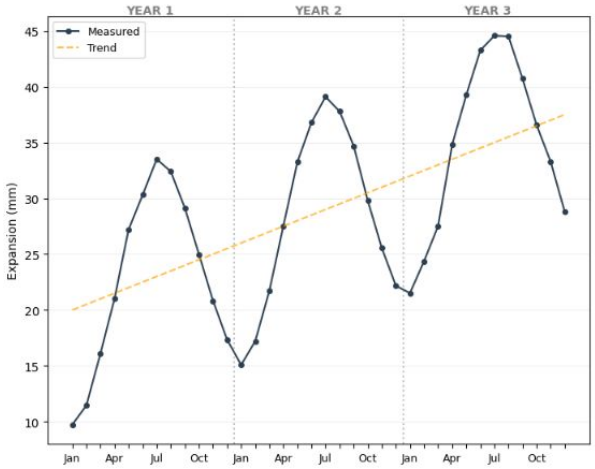
- ***Reveal Trends:*** Filter out recurring patterns to see the true long-term direction.
- ***Forecast Patterns***: Separate predictable seasonality from trend growth for more accurate forecasts.
- ***Detect Anomalies:*** Sudden spikes in residuals highlight sensor errors or unusual events quickly.



Seasonal Decomposition: Additive versus Multiplicative Models

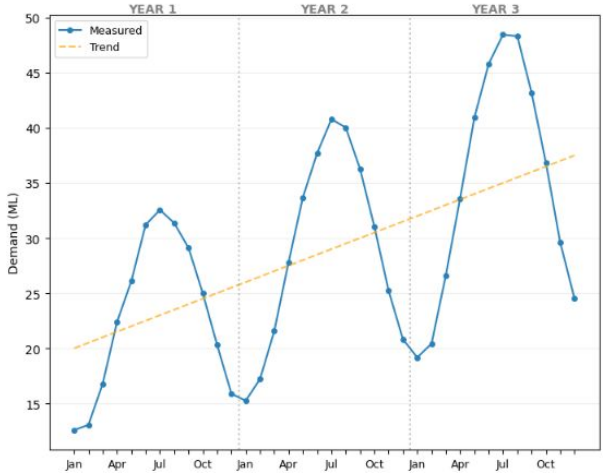
- **Additive:** Seasonal swings stay **constant** in size, regardless of the trend.
- **Example: Bridge Thermal Expansion.** The bridge expands by the same amount every year

Data = Trend + Seasonality + Noise



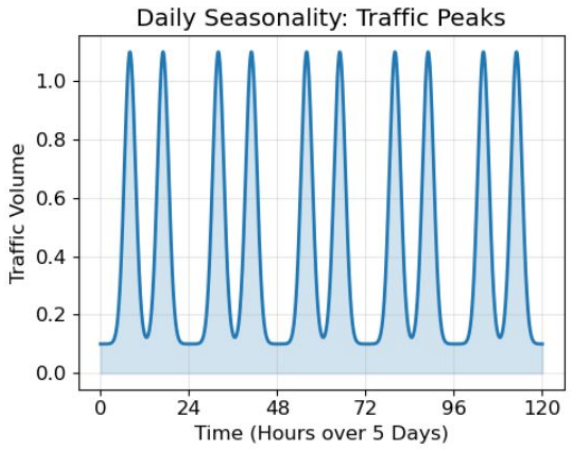
- **Multiplicative:** Seasonal swings **scale** (grow or shrink) in proportion to the trend.
- **Urban Water Demand.** the "Summer vs. Winter" usage gap increases as a city's population grows.

Data = Trend × Seasonality × Noise

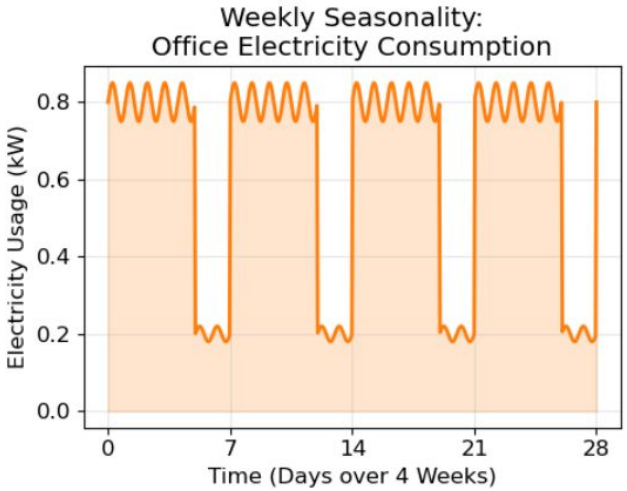


Daily, Weekly and Yearly Seasonality

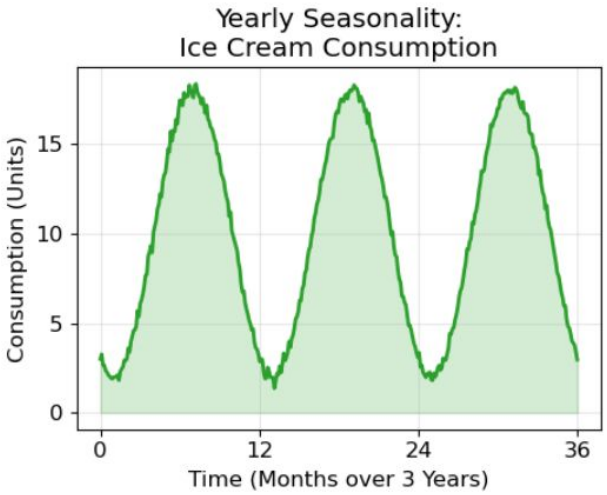
The most common types of seasonality pattern in time series are:



Repeat Every 24 Hours



Repeat Every 7 Days



Repeat Every 12 Months

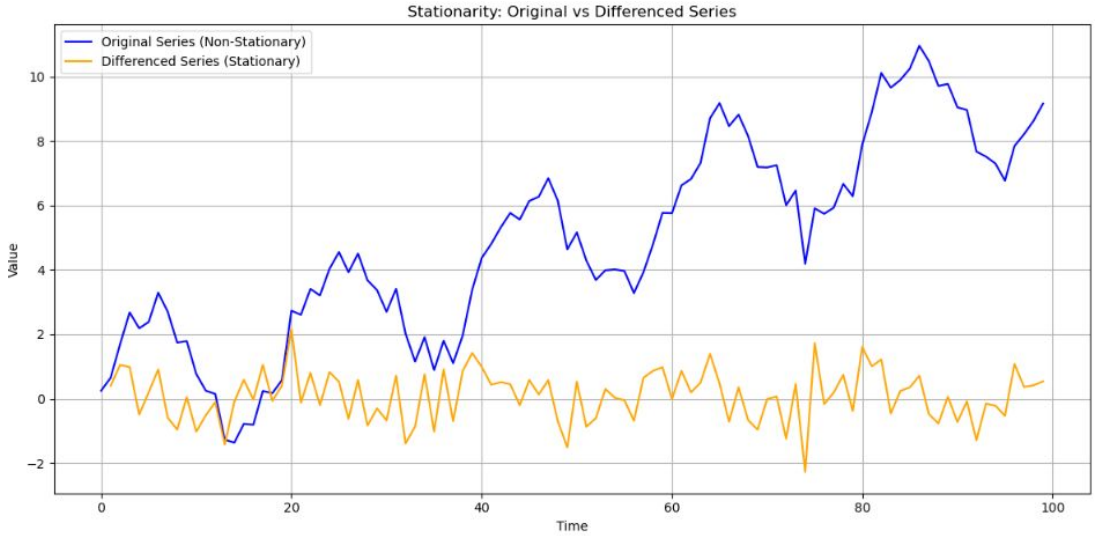
Temporal Dependence

Stationarity versus Non-Stationarity

Stationary Time Series: Statistical properties (mean, variance) don't change over time.

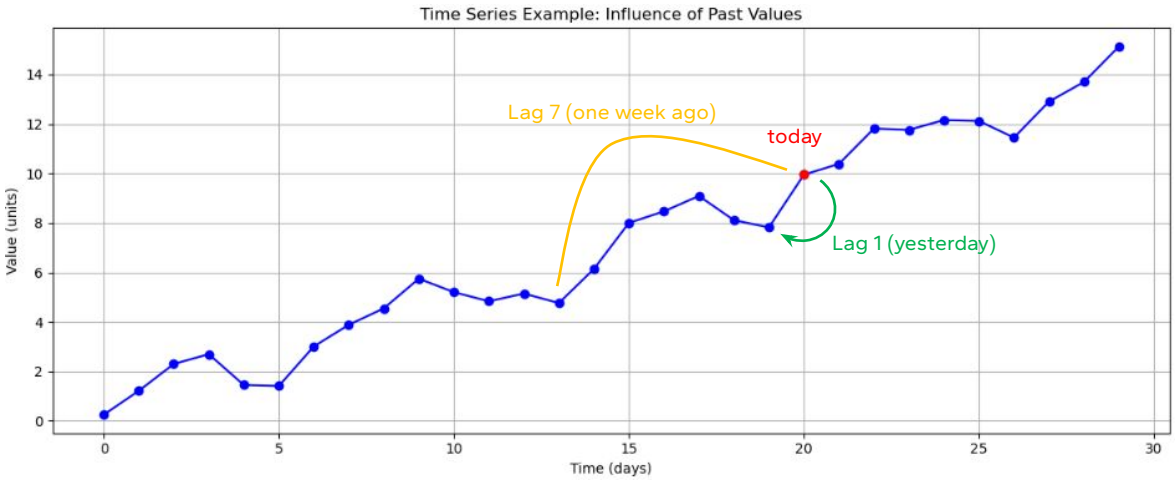
Non-stationary Time Series: Statistical properties change over time (trends, seasonality).

- **Why models care:** Many forecasting models assume stationarity.
- **Spoiler:** 99% of real world time series are *not* stationary
- **Fixing it:** Subtract previous value from current value to remove trend or seasonality.
- **Checking it:** ADF test (Augmented Dickey-Fuller) can verify if a series is stationary.



Why Past Values Matter

- **Memory in time series:** Most real-world processes aren't random; today's value often depends on past values. *Example:* traffic flow, energy usage, or construction progress tends to follow patterns from previous days.
- **Lag concept:** "Lag 1" = yesterday, "Lag 7" = a week ago. Lags capture how past events influence current outcomes
- **Predictive power:** By analyzing past values, we can identify trends, cycles, and repeating patterns, improving forecasts and planning.
- **Practical tip:** Lags are the foundation of many forecasting models (e.g., ARIMA) used in engineering applications



Autocorrelation Function (ACF)

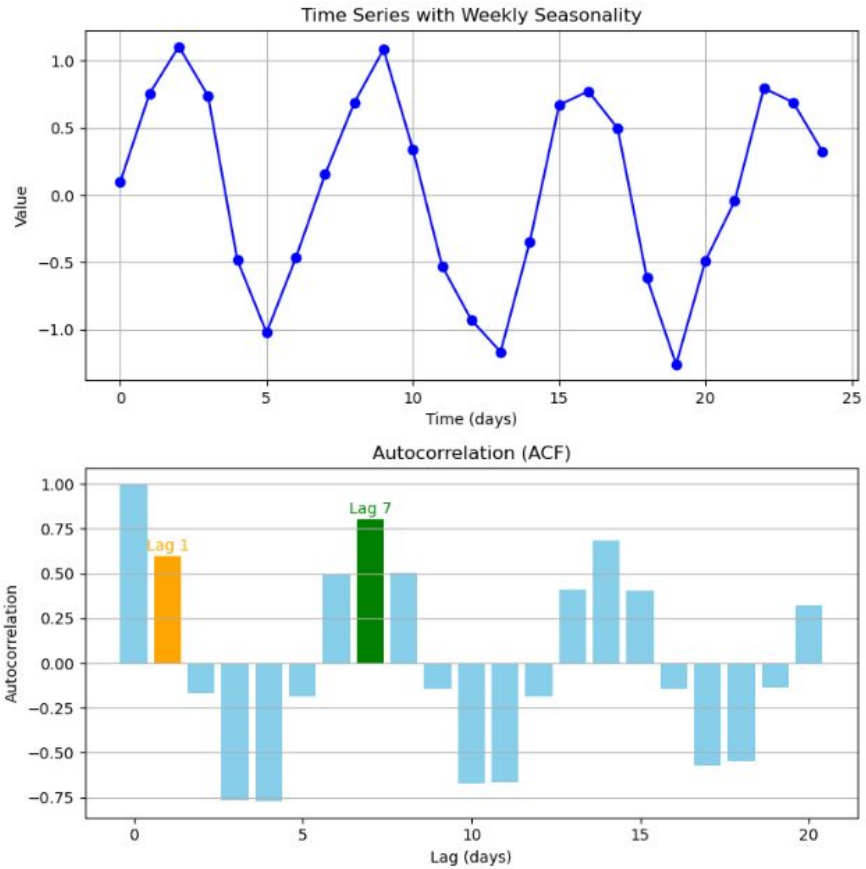
Correlation (basic idea):
Measures how two time series behave relative to each other.

Autocorrelation:
Measures how a time series is correlated with itself at different lags, for example:

- *Lag 1:* Compare today's value with yesterday's value. High ACF value: yesterday strongly influences today.
- *Lag 7:* Compare today's value with the same day last week. High ACF value: weekly patterns.

Why it matters:
Peaks in the ACF denotes strong influence from past data, revealing pattern, cycles, and repeating behavior in the series (daily, weekly, yearly).

What's missing:
ACF includes both *direct* and *indirect* effects, i.e., a week-old value may appear important just because of the influence of the days in between and not because it is directly impacting today



Partial Autocorrelation (PACF)

What PACF measures:

- the direct influence of a past value on the current one, after removing effects of intermediate lags.
- *Example:* Lag 2 (two days ago) may seem correlated with today, but some of that correlation could be due to Lag 1. PACF isolates the direct contribution of Lag 2.

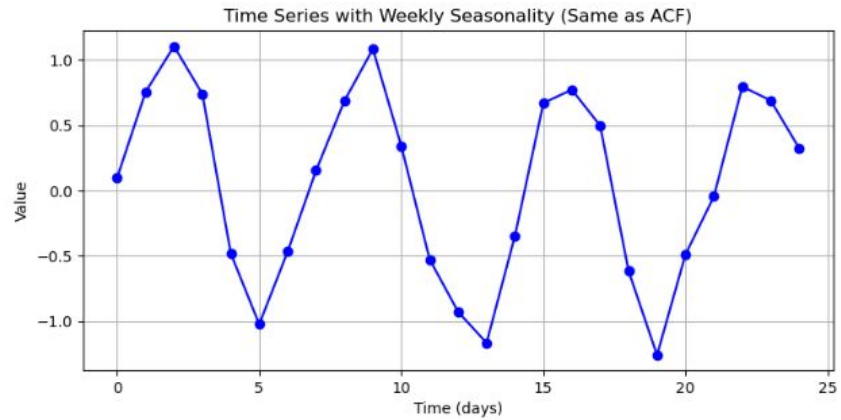
Why it matters:

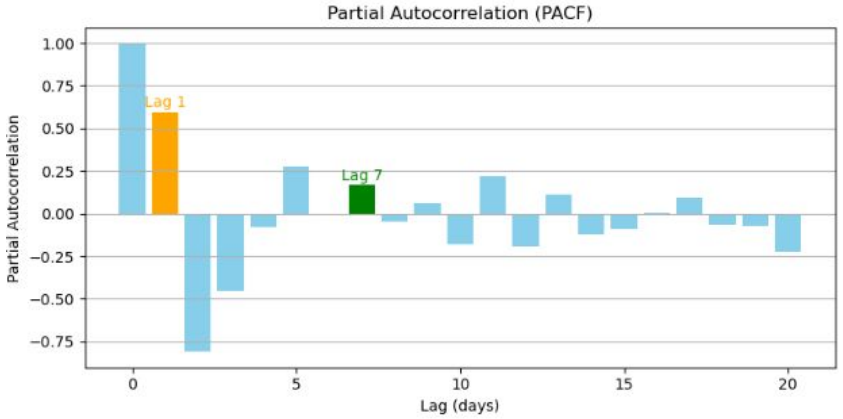
- *Complements ACF:* ACF = total correlation (direct + indirect), PACF = direct only.
- Helps identify how many lags to include in forecasting models (like AR terms in ARIMA).

Link to lag concept:

- Lag 1 (yesterday):shows its direct effect on today.
- Lag 7 (same day last week): shows if there is a direct weekly pattern, ignoring intermediate days.

Practical takeaway: Peaks in the PACF indicate lags that have a strong direct effect on the current value, useful for model selection.





M5 U1 LS2

Predicting the Future: Time Series Forecasting

Forecasting:

Training and Evaluating

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Train-Test Split for Time Series

What does "training a model" mean?

- We feed historical data into a model so it can learn patterns (trend, seasonality, correlations).
- Then we check how well it predicts unseen data.

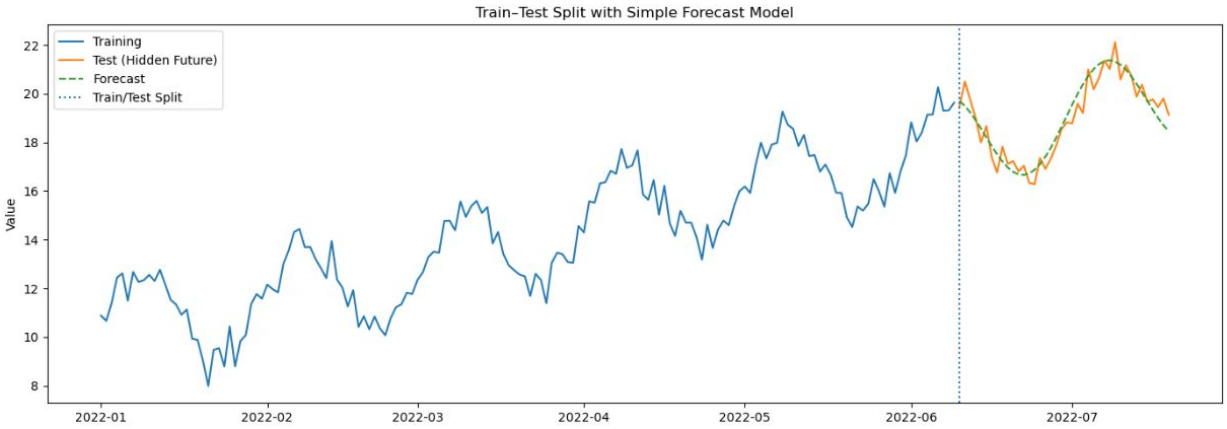
Why do we split the data? To simulate the real world.
Even though we already know the full dataset, we artificially hide the last part to simulate the future:

- Past → used for training
- "Future" → used for testing (evaluation)

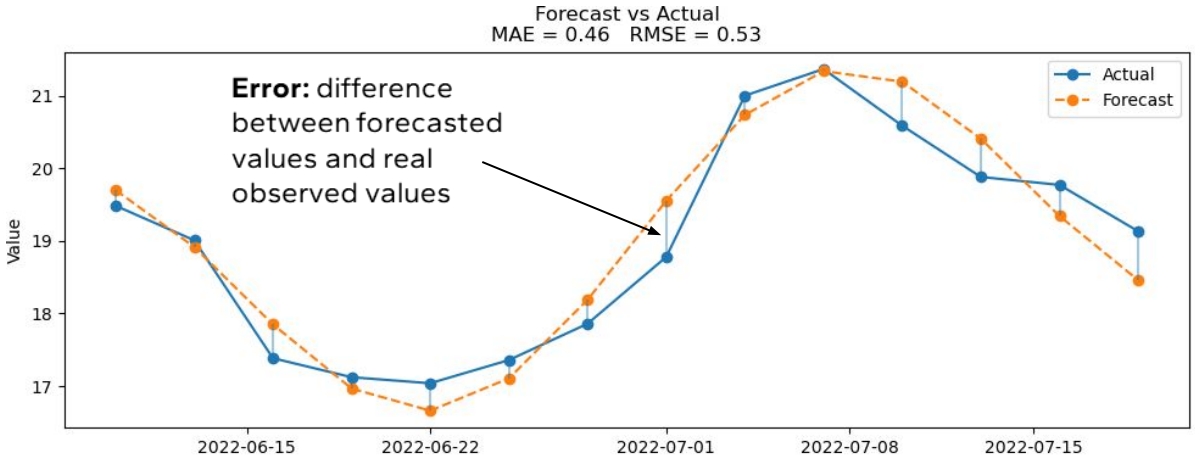
We want to know: How well would this model perform on future data?

Differently from classical machine learning, in time series:
🚫 We **NEVER** shuffle data
✅ We always keep chronological order

How does the final production model work?
We train on all available data, make real forecasts, and only tomorrow we discover how wrong we were



Forecast Accuracy Metrics



Mean Absolute Error (MAE): Average size of the mistakes: “How far off am I, on average?

$$MAE = \frac{1}{N}\sum\left|y_{true} - y_{pred}\right|$$

Root Mean Squared Error (RMSE): Like MAE, but penalizes large errors more: “Big mistakes hurt”.

$$RMSE = \sqrt{\frac{1}{N}\sum\left(y_{true} - y_{pred}\right)^2}$$

Classical Forecasting with ARIMA

ARIMA: The Three Pillars of Time Series

ARIMA is a forecasting model built from three tunable components that work together to capture the 'logic' behind historical data

AR – Autoregressive

The Long-Term Memory (p)

This component assumes that **past values** are the best predictors of the future. It looks for a "momentum" or a repeating pattern in the data points themselves.

I – Integrated

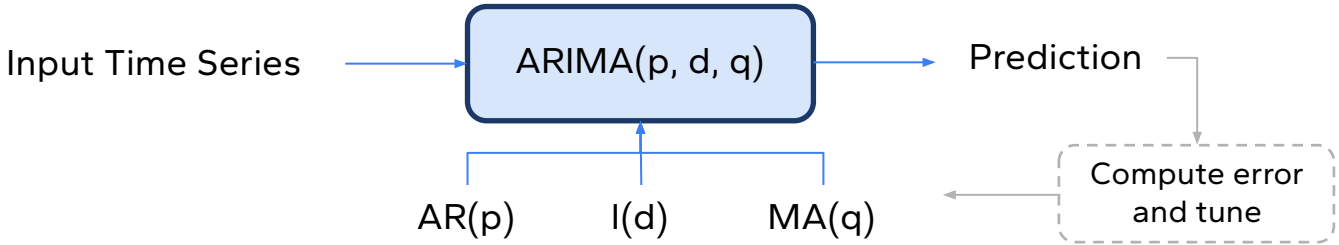
The Stabilizer (d)

This is the "preparation" phase. It uses **Differencing** *d* times (subtracting yesterday from today) to remove trends and make the data **Stationary**.

MA – Moving Average

The Short-Term Shock (q)

This component looks at **past forecasting errors**. It adjusts the forecast based on how much the "shocks" (unexpected spikes) from yesterday still impact today.



Integration (I, *d*): Removing The Trend

Main idea

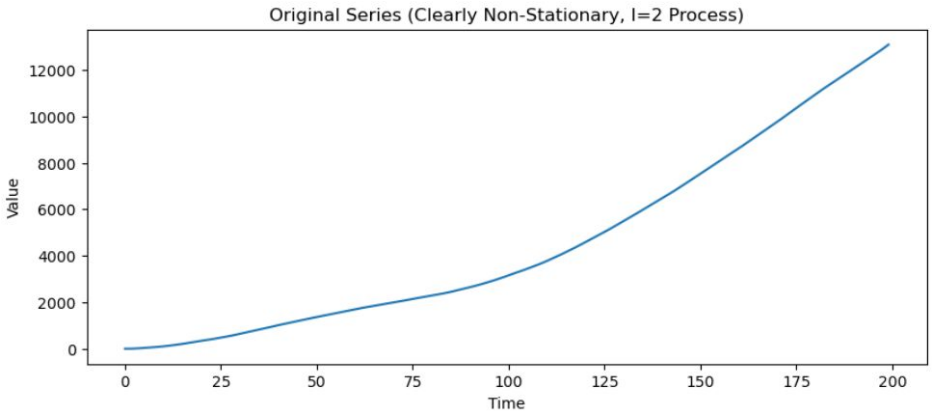
- Before forecasting, we need a stable signal. If the average keeps going up or down, prediction becomes unreliable.
- Differencing = subtracting today minus yesterday. This removes trends.

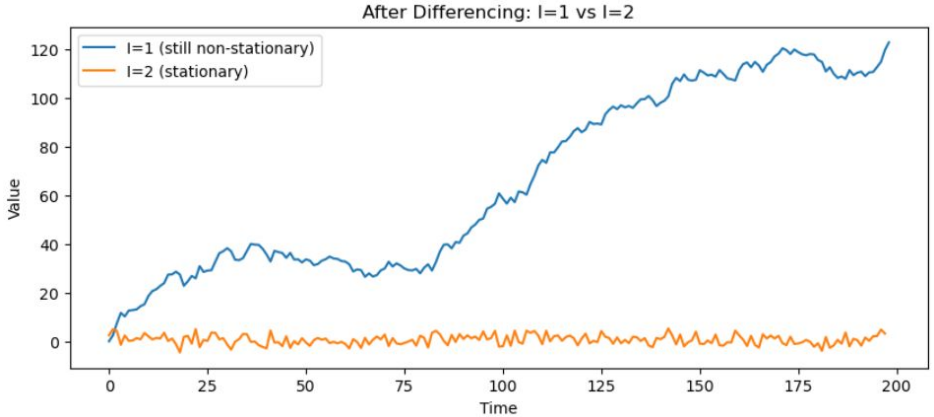
What we see in the picture:

- *Original curve:* clearly increasing, not stable.
- *After 1 difference (I=1):* trend is reduced, but still drifting.
- *After 2 differences (I=2):* now it wiggles around a constant level. This last one is stationary.

Rule of thumb: Keep differencing until the curve looks stable.

Usually:
No trend → d = 0
Simple trend → d = 1
Strong curved trend → d = 2



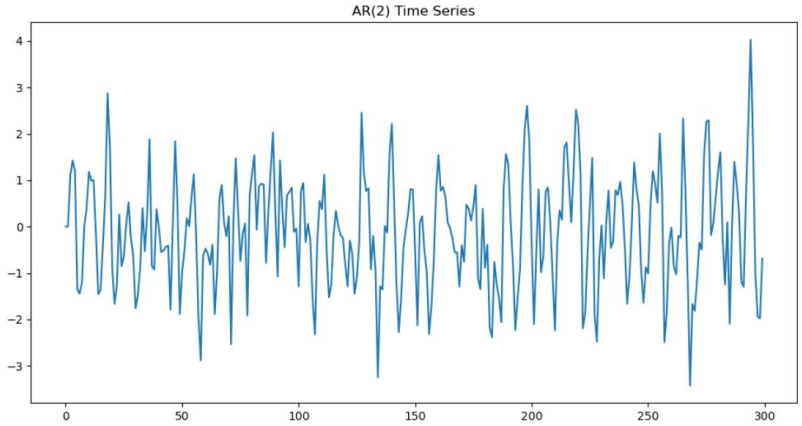


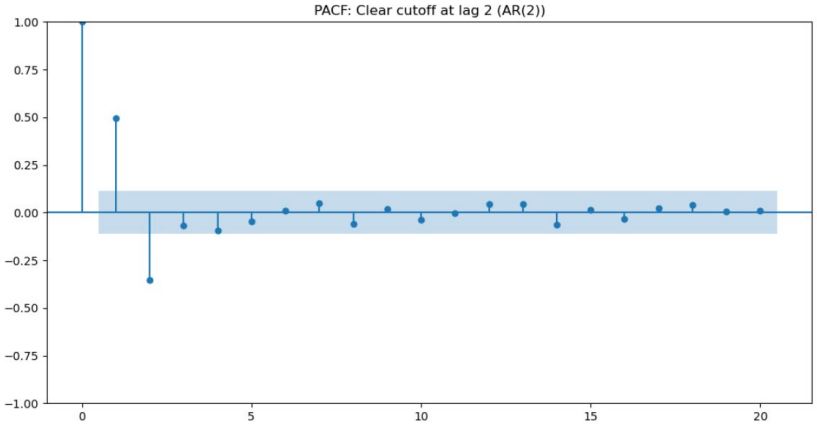
\ Autoregressive (AR, *p*): How Much Does the Past Matter?

Main idea: How many previous values influence the current one?
- Yesterday affects today.
- Maybe also the day before yesterday.
- PACF helps us see how far back this influence goes.

What we see in the picture:
- Strong spikes at lag 1 and lag 2
- After that, almost nothing
- This tells us: Value at time t depends on “t-1” and “t-2”
- Older history does not matter much.

Rule of thumb:
- Look at PACF
- Count how many early spikes stand out.
- Stop when they disappear.





\ Moving Average (MA, q): How Long Do Shocks Last?

Main idea: When something sudden happens, how long does its effect stay in memory?

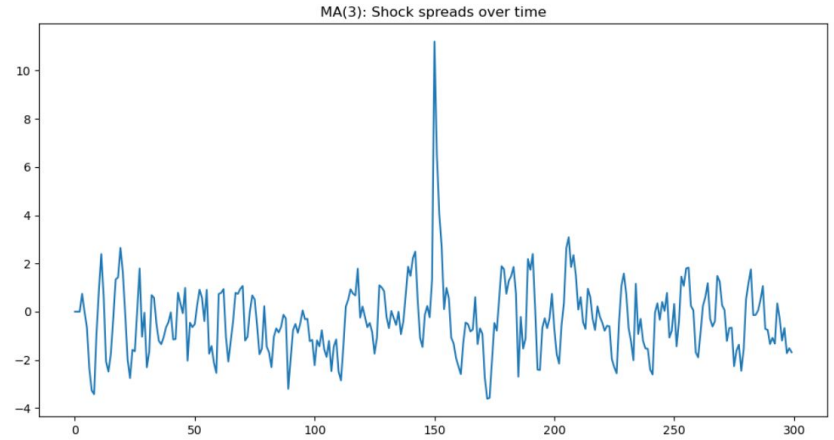
- *Examples:* Sensor glitch, Sudden demand spike, Random disturbance, etc.,
- MA models how these shocks fade away. ACF shows this.

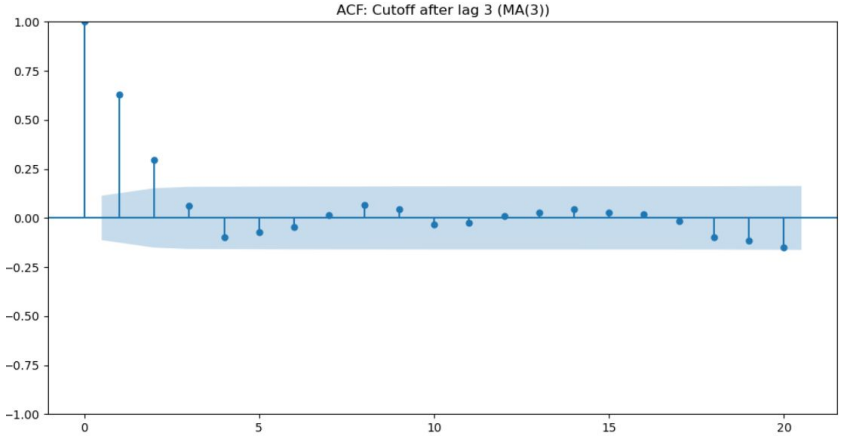
What we see in the picture:

- *Original curve:* shows a clear spike (shock).
- *ACF shows:*
  - *Big correlation at lag 1*
  - *Smaller at lag 2*
  - *Almost gone after that*

Rule of thumb:

- Look at ACF
- Count how many lags are affected by the shock.





ARIMA in Practice: Strengths & Boundaries

The "Reality Check“: ARIMA is a powerhouse for linear patterns, but it requires the data to "behave" before it can provide reliable forecasts.

Strengths & Limitations:

- *The Gold Standard:* Excellent for short-to-medium term forecasting where the past is a strong predictor of the future.
- *The "Stationarity" Barrier:* ARIMA struggles with highly non-stationary series (data that changes its behavior or variance wildly over time). If you can't stabilize the data through differencing, ARIMA's accuracy collapses.
- *Complexity Cap:* Standard ARIMA is "blind" to repeating cycles (like a weekly sales spike or a monthly bill).
- *Difficult to Tune with ACF/PACF/ADF:* in practice we try several combinations of parameters and see which works best.

Beyond the Basics...

Model	The "Upgrade"	Best Used For...
SARIMA	Adds Seasonality	When your data has one clear cycle (e.g., Higher ice cream sales every Summer).
SARIMA**X**	Adds Exogenous Variables	When external factors drive your data. "Exogenous" means outside data, like using Weather or Holidays to help predict Energy Demand.

Modern Forecasting with Prophet

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Meta's Prophet: The Business Forecaster

What is Prophet?
Prophet is a 'plug-and-play' forecasting tool designed by Meta to handle business data that is messy, seasonal, and influenced by humans.

The Working Principle:
- Prophet views a time series as a combination of independent components added (or multiplied) together.
- It then forecasts each part separately and then merge them back together.

The Core Equation: Prophet doesn't look at "lags" like ARIMA; it calculates a curve for each of these:

$$TimeSeries = Trend + Seasonality + Holidays + Error$$

Two Ways to Mix the Ingredients:
- *Additive (Standard):* Components are simply added. Seasonal "swings" stay the same size even if the trend grows.
- *Multiplicative (Growth):* Components are multiplied. Seasonal "swings" grow or shrink in proportion to the trend.

Prophet in Practice: The Three Components

Prophet is designed to be simple to use and tune. By tuning these parameters, you define the behavior of the forecast.

Seasonality

The Rhythm (s)

Handles daily, weekly, and yearly cycles. It can model multiple cycles at once.

Trend

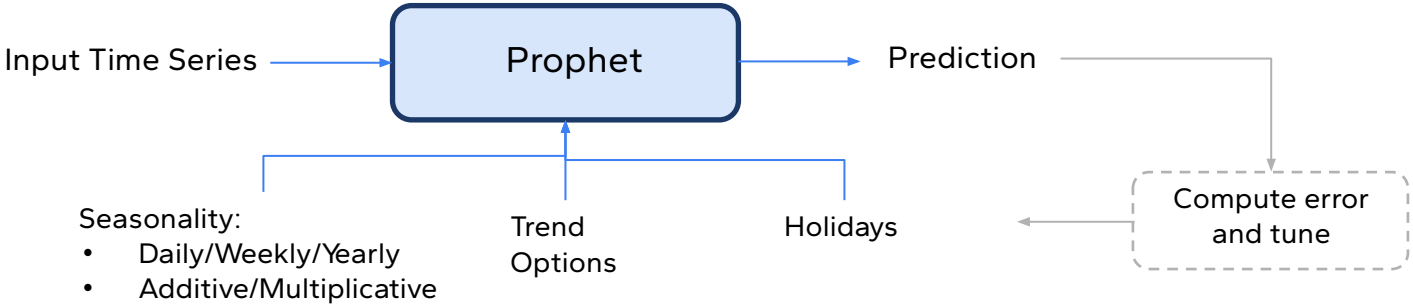
The Direction (g)

Handles non-periodic changes. Can detect where growth speed shifted (e.g., a new market entry).

Holidays

The Shocks (h)

Handles "One-off" events. You provide a list of dates, and Prophet assigns them a specific "effect" so they don't skew the trend.



Prophet: Strengths & Limitations

Why it's a Business Favorite:

- *Robustness:* Handles missing data and massive outliers without crashing or needing manual "gap-filling.“
- *Domain Knowledge:* Tuning is done via business logic (e.g., "Add a monthly cycle") rather than abstract statistical tests.
- *Automation:* Designed to produce "good enough" forecasts for thousands of items quickly.

The Boundaries:

- *Precision:* For extremely clean, high-frequency data (like engine vibrations), ARIMA is often more precise.
- *Simplicity:* It assumes components are independent. If your "Holiday" effect changes based on the "Trend" in a complex, non-linear way, Prophet might struggle.
- *Stationarity:* Unlike ARIMA, it doesn't care about stationarity, which is a strength, but it means it can sometimes "hallucinate" a trend that isn't really there.

Choosing the Right Tool for the Task: ARIMA vs. Prophet

ARIMA is a rigorous statistical framework, while Prophet is a pragmatic tool designed for business operations.

Feature	ARIMA (The Scientist)	Prophet (The Practitioner)
Foundation	**Statistical Formalism:** Grounded in stochastic theory; mathematically precise but rigid.	**Curve Fitting:** A "Generative Additive Model" that treats forecasting like a drawing exercise.
Goal	**Optimal Accuracy:** Aims for the "Best Linear Unbiased Estimator" through expert tuning.	**"Good Enough" at Scale:** Designed to provide reliable forecasts for thousands of items quickly.
Error Handling	**Mathematical:** Actively models the correlation of residuals (MA component) to refine predictions.	**Pragmatic:** Discards residuals as random noise. Focuses only on Trend, Seasonality, and Holidays.
User Profile	Requires **Statistical Expertise** to handle stationarity, lags, and significance tests.	Accessible to **Business Analysts** who know the domain but aren't time-series experts.

Deep Learning Forecasting (Optional)

Neural Networks for Time Series: The Idea (RNN)

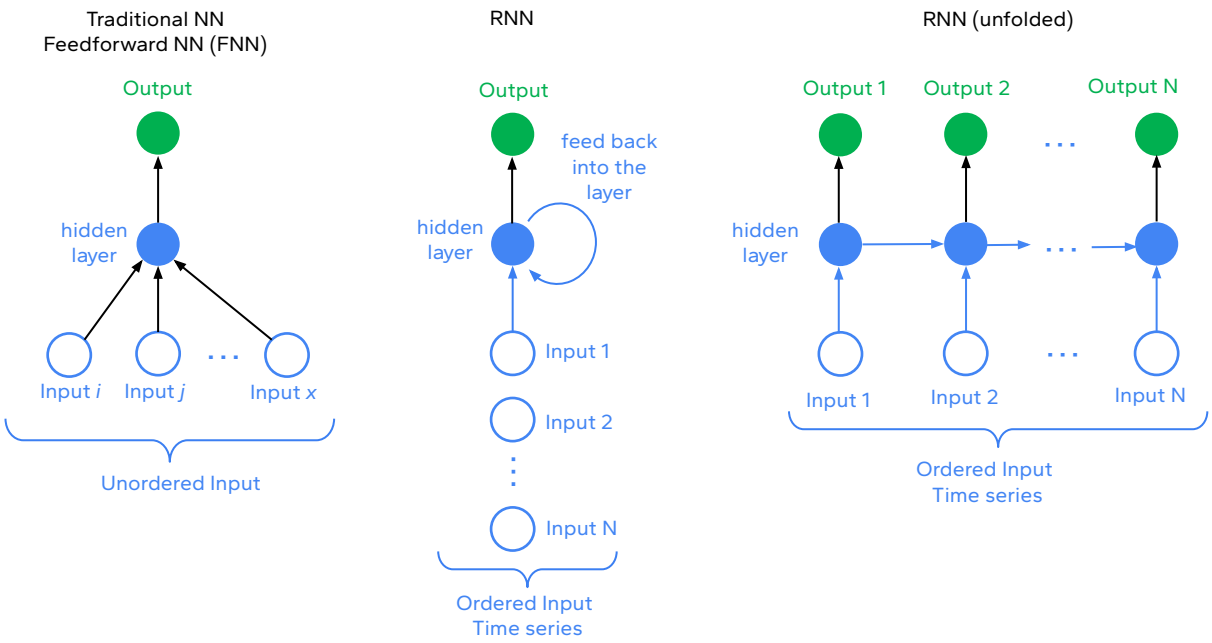
Core Intuition
A Recurrent Neural Network (RNN) is a Neural Network (NN) designed for sequential data, like time series.

Difference from traditional NN

- In traditional NNs (a.k.a. feedforward), information flows forward only. *RNNs reuse the same hidden state over time*.
- Past information is carried forward and mixed with the current input, allowing the model to learn temporal correlations.

At Each Time Step:

- Current value comes in
- Past memory is updated
- A prediction is produced



RNN vs Classical Forecasting

Supervised Learning

RNNs are *supervised* models:

- We give historical inputs *and* known outputs
- The network learns by minimizing prediction errors

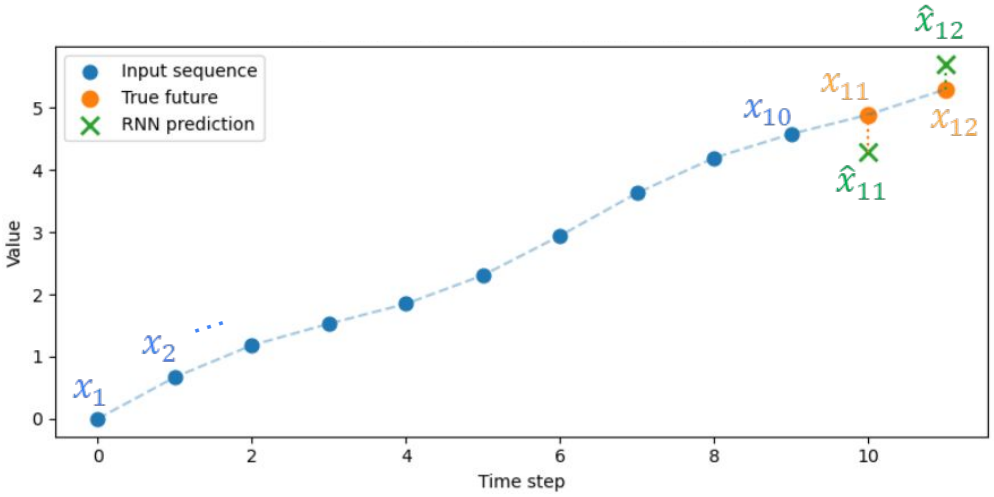
Difference from Statistical Forecasting:

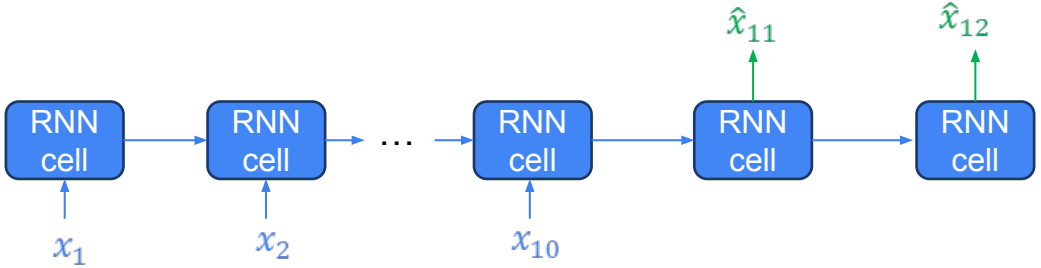
ARIMA / Prophet:

- Statistical models
- Explicit assumptions (trend, seasonality)

RNN:

- Learns patterns automatically
- No predefined structure
- Data-driven





Long Short-Term Memory (LSTM)

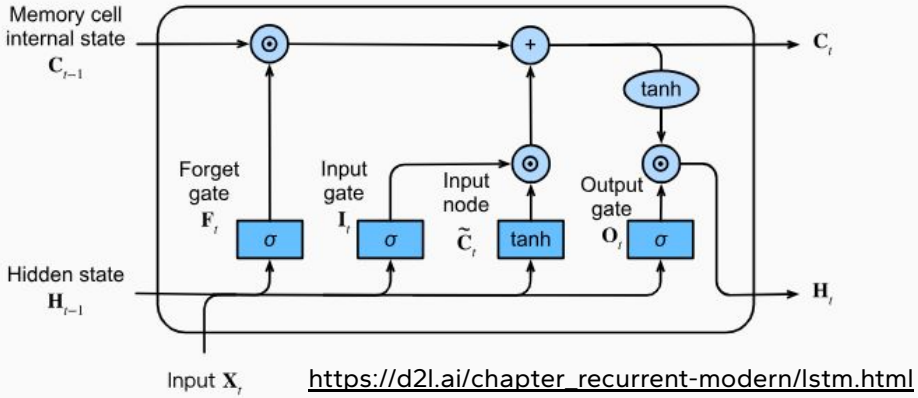
Why LSTM? When Should You Use Deep Learning?

Problem with basic RNNs
- They struggle to remember information far in the past.
- Important patterns can be forgotten.

LSTM is a special RNN designed to:
- Remember long-term patterns
- Decide what to keep and what to forget
- Handle complex temporal relationships

When does LSTM make sense?
- Very large datasets
- Highly complex patterns
- Many interacting variables
- Classical models underperform

Otherwise: Start with ARIMA or Prophet.



https://d2l.ai/chapter_recurrent-modern/lstm.html

Key Takeaways

Key Takeaways

What is a Time Series?

- A sequence of measurements ordered in time (energy, traffic, costs, temperature, heart rate...). *Order matters*.

What is Forecasting?

- Using past data to estimate future values.
- *Goal:* support better decisions, not predict perfectly. Forecasting is about being less wrong than guessing.

Time Series Have Structure

- Most series contain: *Trend* (long-term direction), *Seasonality* (repeating patterns) and *Noise* (random variation).
- We analyze patterns using *Seasonal Decomposition*, *Autocorrelation* (ACF / PACF) and *Stationarity* (sometimes with differencing).

Forecasting Models: Learning patterns and project them forward

- *ARIMA:* Classical statistical model; Works well on *stationary* data; Requires *manual tuning* and interpretation
- *Prophet:* Modern, automatic approach; Handles trend + seasonality out of the box; Robust to missing data and holidays; Easier to use in practice.

Time series forecasting turns measurements into expectations, and expectations into decisions.

Thank You!

Q&A

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